Aanand Ajith

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.46
Kuriakose Elias English Medium School	Class XII (ISC)	2024	95.60%
L'ecole Chempaka	Class X (ICSE)	2022	97.83%

SKILLS

Programming Languages : Python | SQL | Java | C

Software & Fundamentals: Data Structures & Algorithms (DSA) | Object-Oriented Programming (OOP) | Operating Systems (OS)

Libraries: Pandas | NumPy | scikit-learn | Matplotlib

Developer Tools: GitHub | Jupyter Notebook | Google Colab | VS Code

Domains: Machine Learning | Data Science | Predictive Analytics | Statistical Modeling | Risk Analytics

PROJECTS

Retail Credit Scorecard Validation Engine | Python, scikit-learn, OpenML

- Built a custom **Python validation pipeline** to audit credit scorecards on **1,000+ real loan records**.
- Ran **Out-of-Time (OOT) back-testing** on an 80/20 chronological split to simulate future applicant data.
- Handled missing data by applying **median imputation** across key continuous financial features.
- Evaluated class separation, reaching a **0.38 KS Statistic** (exceeding the **0.30 industry benchmark**).
- Achieved strong ranking discrimination with an **OOT AUC of 0.74** and a **Gini of 0.48**.
- Monitored overall score drift using **Population Stability Index (PSI < 0.08)** to verify stability.
- Calculated **Characteristic Stability Index (CSI)** to pinpoint drift in specific borrower features like income.

Challenger Model Benchmarking & Data Leakage Audit | Python, XGBoost, SQL

- Benchmarked a baseline **Logistic Regression scorecard** against an **XGBoost challenger model**.
- Audited feature creation dates to detect **data leakage** from post-approval repayment variables.
- Identified **30+ Days Past Due status** as a leaked feature and removed it to prevent false metric inflation.
- Evaluated model performance using **Precision, Recall, and Confusion Matrices** across both models.
- Weighed the challenger's **predictive lift** against strict regulatory **Adverse Action explainability rules**.
- Summarized findings into a **Model Risk Governance report** recommending final deployment decisions.

SaaS Customer Churn Analysis | SQL, Excel, Power BI

- Analyzed **500 customer accounts** using **SQL and Excel** to spot cancellation patterns.
- Grouped account data across **industry types**, **subscription tiers**, and **user segments**.
- Uncovered an overall **22% churn rate** (110 canceled accounts) and mapped retention drop-off points.
- Calculated key retention metrics across cohorts to highlight at-risk customer groups.
- Built an interactive **Power BI dashboard** with KPI cards to present actionable retention strategies.

Generalized Planar Mechanism Solver | **Link**

- Engineered an **interactive web application** to draw and simulate **arbitrary planar mechanisms** with unrestricted links and sliders.
- Designed a **generalized object-oriented kinematic solver** utilizing the **Newton-Raphson method** and **numerical Jacobian**.
- Eliminated the requirement for **mechanism-specific analytical equations**, ensuring flexible configuration.
- Validated framework scalability across **complex linkage systems** by testing on mechanisms with **50+ links**.
- Expanding solver capabilities to support **velocity, acceleration, and dynamic analysis** via **Jacobian-based formulations**.

EXTRACURRICULAR

- Participant in the **Trivandrum Cricket League**.
- Participant in **Milan '24(Inter hostel competition of IITH)** in **cricket and football**. Runners up in football.
- Proficient in **badminton and volleyball.**
- Participated in high school/higher secondary school level **athletic meets**. Won several medals as a sprinter.

POSITIONS OF RESPONSIBILITY

Member of **Aeolus**, **the Racing Club of IITH**

- Evaluated multiple airfoil designs by analyzing performance data to identify the profile with the highest aerodynamic efficiency for the vehicle.
- Analyzed geometric constraints and flow data for the undertray and diffuser section to determine the optimal fit and integration with the car's chassis.

RELEVANT COURSES

- **Computer Science:** Introduction to Programming
- **Mathematics:** Introduction to Statistics, Introduction to Probability, Elementary Linear Algebra, Calculus I, Calculus II
- **Liberal Arts:** Introduction to Entrepreneurship, Communication Skills